Brian Wonjun Lee

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Education

University of Pennsylvania — Philadelphia, PA
MSE in Computer Graphics & Game Technology (CGGT), Submatriculation — *Expected May 2028*
BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars — *Expected May 2027*
Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Experience

Software Engineer Intern, Aleph Lab (Y Combinator) — *June 2026 – Aug 2026 · San Francisco (Remote)*

- **Changed what every user opens the app to** — restructured the information architecture so speaking practice got its own destination instead of a banner buried under the home feed, and the home grid doubled the modes a child sees at a glance; **live to 100% of users** behind a one-click kill switch, the team's biggest engagement bet of the quarter
- **Made a full rebrand cost users nothing to relearn** — shipping structure first meant the layout was already familiar when the skin flipped, then a **25→50→100%** ladder held across real class cycles with continuity on, so a partial rollout could never change a child's design mid-session; **13 PRs**, a **3,748-test** gate, byte-identical flag-off, and **3.3 MB** lighter to download
- **Turned a channel that went dark whenever nobody approved a send into one that runs itself** — proposing, clearing its own guardrails and reaching **91% of targeted devices** with **no human in the loop**
- **Produced the first revenue the win-back ever generated**, and reached **~2.7× more users** after proving the eligibility logic had excluded most of its intended audience
- **Made a children's app safe from notification injection** — anyone with a user ID could push arbitrary content until I bearer-authed four public endpoints, **zero caller changes**
- **Stopped the app reporting failures as facts** — a failed request had been telling parents their child spoke in no classes; split error from empty across the growth screens, and replaced the blank white crash screen with a branded recovery in Korean and English

Software Engineer Intern, BitMango (Puzzle1 Studio) — *June 2025 – Aug 2025 · Pangyo, South Korea*

- **Kept 50+ live Unity mobile titles shipping** across a summer of releases, holding UI/UX and gameplay stable
- **Cleared 100+ QA-reported bugs into production**, in Unity prefabs and C# across a portfolio no single engineer owned
- **Kept the portfolio compliant and in the stores** through a platform requirement change, upgrading SDKs fleet-wide

Project Lead & Product Designer, Penn Spark — *Jan 2025 – Present · Philadelphia, PA*

- **Shipped client-facing 0-to-1 products** leading a cross-functional team of **15+** designers and developers

Database Engineer Intern, IT-Farm — *June 2022 – Aug 2022 · Seongnam, South Korea*

- **Made semiconductor data from 20+ client companies queryable** — including Samsung, SK and LG — by designing a SQL database prototype managing **1M+ entities** in Oracle and SQLite
- **Cut retrieval time across large-scale datasets** by restructuring and optimizing the queries the team relied on

Projects

Capsule | *React, Three.js (R3F), Blender* — *Jan 2025 – Apr 2025*

- **Made memories something you walk through rather than scroll** — the **3D gallery** in React Three Fiber for a collaborative time-capsule app (team of 4), with direct-manipulation placement
- **Owned how the product felt to use** across the create → contribute → unlock flow

Mini Maya | *C++, OpenGL, Qt* — *Feb 2026 – Mar 2026*

- **Made arbitrary meshes editable in place** — a **half-edge structure** giving topology-aware traversal, plus an OBJ parser accepting arbitrary n-gons, rendered via OpenGL VBOs
- **Turned coarse geometry into smooth surfaces** with **Catmull–Clark subdivision** and topology ops (split, extrusion, triangulation) in a Qt editor

Mini Minecraft | *C++, OpenGL, GLSL* — *Apr 2026*

- **Made a from-scratch voxel world read as varied rather than noisy** — **7-biome procedural generation** (Perlin/FBM/Voronoi) and **3D Perlin caves** in a C++/OpenGL engine (team of 3)
- **Made that world legible** — the GLSL pipeline I owned: PCF shadows, screen-space reflections, vertex AO, day-night cycle

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C++, C#, SQL, HTML/CSS, OCaml
Frameworks & Libraries: React, Next.js, Node.js, Astro, Tailwind CSS, Three.js, React Three Fiber, JUnit
Developer Tools: Git, GitLab, AWS, MongoDB, SQLite, Oracle, Figma, VS Code, IntelliJ
Graphics & 3D: OpenGL, GLSL, Unreal Engine 5, Unity, Maya, Blender